Chapter 11

Property Taxes

A tax on the value of privately owned property is called a property tax. An important difference compared to previous taxes is that the government sets the tax base and the tax rate. Determining the base and setting the rate are topics covered in this chapter as well as measures related to equity and policies to lessen the burden on particular groups of taxpayers.

In theory, property taxes are taxes on wealth (e.g., real estate, personal property). The inclusion of all property is administratively and politically not feasible and thus, property taxes are usually limited to agricultural land, commercial and industrial property, and residential homes. Some properties are exempt (e.g., public properties, cemeteries, churches, colleges, nonprofit hospitals) as long as they are used for the intended purpose. Property taxes are the most important revenue sources for local governments in the United States. In 2024, property tax revenue as a share of general revenue from own sources (taxes in this case) at the local level was 70.1% (see U.S. Census Annual Survey of State and Local Government Finances for more information). Property taxes represent an important economic basis for local autonomy.

The administrative steps for collecting property taxes begin with the identification and registration of property that is subject to taxation based on ownership. In a second step, the properties' values are quantified to establish the tax base. The tax rate is set based on local public service requirements.

11.1 Property Value Determination

The goal of property value assessment is to provide an accurate and current estimate of the value for each property. The accuracy of these assessments directly impacts the (perceived) fairness of property taxes. Achieving higher accuracy typically involves more frequent and regular assessments, ideally on an annual basis. There are three types of property values:

1. **Market value** assumes full information and properly functioning real estate markets.
2. **Appraised value** is determined by a professional appraiser and is relevant in the sale process (e.g., for mortgage lending purposes).
3. **Assessed value** is used for property tax purposes and is often set by policy as a fraction (i.e., between 0 and 1) of the market value. The assessed value may differ from market and appraised values. Property value assessments are typically conducted at the local level and rarely at the state level. While market and appraised values are defensible due to their basis in actual transactions, assessed values are primarily determined as part of the tax policy.

There are three main methods used to assess property values: (1) comparable sales approach, (2) cost approach, and (3) income approach. The comparable sales approach is frequently used for residential properties by leveraging market transaction data. Since there is a sufficient number of transactions per year, statistical methods insure a relatively accurate property value determination. The cost approach is more suitable for industrial and utility properties. It involves determining either the reproduction cost (i.e., the cost to construct an identical property) or the replacement cost (i.e., the cost to build a similar property using modern technology). Of course, this may be more difficult if you consider for example the value of a unique facility such as an car manufacturing plant. The income approach is commonly used for income-generating properties such as apartment complexes. It determines the annual profit from the property and uses the net present value (NPV) to determine its current value. In the case of property valuation, the revenue is assumed to be received in perpetuity.

Let us consider the example of a 10-unit apartment complex that rents for $1,500 per month. The expense rate as a percent of income is 54% and the vacancy rate is 6%. The discount rate used for the NPV calculation is 5%. Given those information, we can calculate the following components:

- Gross income: $\$1,500 \cdot 10 \cdot 12 \cdot (1 - 0.06) = \$169,200$
- Expenses: $\$169,200 \cdot 0.54 = \$91,368$
- Net income per year: $\$169,200 - \$91,368 = \$71,832$
- Present value: $\$71,832/0.05 = \$1,436,640$

The assessment ratio (AR) is the assessed value (AV) over market value (MV).

$$AR = \frac{AV}{MV}$$

The ratio is set by state finance laws to differentiate classes of property such as residential, agricultural, commercial, and industrial. There are lower rates for residential properties than for commercial and industrial property. The rationale for property classifications are the perception of fairness and slow urban sprawl by taxing farm property at lower rates.

The remaining question is with regard to the frequency of property value assessment. There are four primary types of assessment cycles used for property taxes.

In the cyclical assessment approach, all properties are assessed during a designated assessment year. Once a property is assessed, its value remains unchanged until the next scheduled assessment, unless significant changes are made to the property that would warrant a reassessment. In a segmental assessment, only a fraction of all properties are reassessed each year. For example, one-third of all properties might be assessed every three years. In the annual assessment, each characteristic of a property is assigned a value, and these values are adjusted based on market trends in the jurisdiction. Finally, under assessment on sale, a property is reassessed when it is sold. This method can lead to significant inequities, both horizontally (between similar properties) and vertically (across different property values), as properties that do not change ownership may not be reassessed for many years.

11.2 Property Tax Rates

The setting of the property tax rate occurs as part of the annual budget process. The simplest rate setting can be expressed with the following formula:

$$r = \frac{E - NPR}{AV}$$

where r represents the tax rate determined by the budget expenditure (E), non-property tax revenue (NPR), and total assessed values (AV). By setting property tax rates during the budgeting process, public managers and policy makers are able to balance the budget and avoid borrowing. However, fluctuations in tax rates might lead to political unpopularity of the tax.

With regard to property taxes, the mill rate refers to the property tax rate expressed in mills, meaning one dollar of tax for every one thousand dollars of assessed value. The relationship between the mill rate (m) and the tax rate (r) is given by $r = m/1000$. For example, if the mill rate is $m = 7$, then the tax rate is $r = 0.007 = 0.7\%$. Thus, tax revenue can be expressed in terms of either r or m:

$$TR = r \cdot AV$$
$$TR = m \cdot AV/1000$$

As an example, assume taxable real property valued at \$220 million with an assessment ratio of 0.5. Exemptions reduce the assessed value by \$3 million, and the total budget expenditures amounts to \$3.5 million, with non-property tax revenue contributing \$0.75 million. The calculation of the statutory tax rate involves multiple steps. First, the gross assessed value is $220,000,000 \times 50\% = 110,000,000$ and the net assessed value is $110,000,000 - 3,000,000 = 107,000,000$. The statutory tax rate is calculated as:

$$r = \frac{3,500,000 - 750,000}{107,000,000} = 0.0257$$

This corresponds to a mill rate of $m = 25.7$.



Figure 11.1: Example distribution of assessed values (in paranthesis) for six parcels covering two school districts.

11.3 Public Services and Rate Setting

We mentioned before that property taxes are an important source for local government funding. In this section, we determine how property tax rates are set in a more complex manner than presented in the previous chapter by linking them to expenditures. In this example, we have six parcels labeled A through F and four public services: (1) two school districts, (2) three Waste water treatment districts, (3) two fire districts, and (4) three library districts. Figure 11.1 illustrate the distribution of school district across the parcels with the number in parenthesis representing the assessed value. For the other distribution of public service districts and parcels, consider Table 11.1. The tax rate required for each district to cover its expenditures is equivalent to dividing the expenditures by the tax base of the corresponding district.

And thus, the rate by parcel is the sum of the rates depending in which district the parcel is located. For example, parcel A is located in districts requiring a rate of $0.1 + 0.08 + 0.02 + 0.05571 = 0.25571$ (Table 11.2). To determine the taxes paid by parcel A, we have to multiply the applicable rate of 0.25571 to A's property value of 100. The tax bill which needs to be paid by property a is $25.57.

Service District	Parcels	Expenditure	Tax Base	Rate
School 1	A+B+C	20	350	0.0571
School 2	D+E+F	10	450	0.0222
Waste Water 1	A+D	5	250	0.0200
Waste Water 2	B+E	5	250	0.0200
Waste Water 3	C+F	5	300	0.0167
Fire 1	A+B+D+E	40	500	0.0800
Fire 2	C+F	20	300	0.0667
Library 1	B+D+E+F	60	500	0.1200
Library 2	A	10	100	0.1000
Library 3	C	10	200	0.0500

Table 11.1: Public service districts and parcel served as well as expenditure and tax base per district. The tax rates are obtained by dividing the expenditures per district by the tax base in the corresponding district.

11.4 Property Taxes: Metrics and Relief

Two commonly used equity-related measures are the coefficient of dispersion (COD) and the price-related differential (PRD). The COD serves as a dispersion measure with regard to the assessed values and serves as a measure of horizontal equity. It reflects the average deviation around the median assessment ratio as a percentage of the median and measures how uniformly properties are assessed relative to their market values. A lower COD generally means more consistent assessments. This measure indicates how closely assessed values are clustered around the median, with less variation being a positive sign. In terms of interpretation, a smaller COD is generally viewed as positive. The International Association of Assessing Officers publishes the Standard on Ratio Studies which contains COD standards, which are acceptable ranges. For residential improved homes, the COD range is 5–15 and for newer homes, it is 5–10. For commercial properties and vacant land, the COD can range from 5–30. Consider the example

Parcel	Value	School	Waste	Fire	Library	Rate	Tax Bill
A	100	1	1	1	2	0.2557	25.57
B	50	1	2	1	1	0.2771	13.86
C	200	1	3	2	3	0.1905	38.10
D	150	2	1	1	1	0.2422	36.33
E	200	2	2	1	1	0.2422	48.44
F	100	2	3	2	1	0.2256	22.56

Table 11.2: Parcel values, service districts, and tax bills based on the rates required for each district to provide its services.

Property	A	B	C
Market value	$40,000	$60,000	$100,000
Assessed value	$25,000	$30,000	$40,000
Assessment ratio	0.625	0.500	0.400
Difference from median	0.125	0.000	0.100

Table 11.3: Example table of properties to calcualte the coefficient of dispersion

properties in Table 11.3. The COD is calculated as follows:

$$COD = \frac{100}{\tilde{r}} \cdot \frac{1}{n} \sum_{i=1}^{n} |r_i - \tilde{r}|$$

where r_i is the assessed value divided by the market (sale) value for property i, and $\tilde{r}$ is the median ratio. The median assessment ratio for our example is 0.5 given the ratios of 0.4, 0.5, and 0.625. The average difference from the median is $(0.125 + 0 + 0.01)/3 = 0.075$ and thus, the COD is $0.075/0.5 \cdot 100 = 15$. Put differently, the average of assessment ratios are 15% from the median.

The Price Related Differential (PRD) is a measurement of tax progressiveness. A $PRD = 1$ indicates a proportional tax whereas A $PRD > 1$ is a sign of a regressive tax meaning there is an under-assessment of high value parcels relative to low value parcels. The opposite case is a $PRD < 1$ indicating a progressive tax, which over-assesses high value parcels relative to low value parcels. The PRD is calculated as follows:

$$PRD = \frac{1/N \cdot \sum_i A_i/M_i}{\sum_i A_i / \sum_i M_i}$$

where A_i is the assessed value of property i and M_i is the market value of property i. Let us consider the example with five properties as presented in Table 11.4.

	Progressive		Regressive	
Market Value	A_i	A_i/ M_i	A_i	A_i/ M_i
$50,000	$35,000	0.70	$47,500	0.95
$75,000	$60,000	0.80	$67,500	0.90
$100,000	$85,000	0.85	$85,000	0.85
$150,000	$135,000	0.90	$120,000	0.80
$250,000	$237,500	0.95	$175,000	0.70

Table 11.4: Progressive and regressive assessment patterns using the PRD. Under the progressive tax regime, the PRD is equal to 0.95 whereas under the regressive tax regime, it is equal to 1.061.

	Property A	Property B
Market value	$100,000	$500,000
Assessment ratio	70%	70%
Assessed value	$70,000	$350,000
Exemption	$30,000	$30,000
Taxed value	$40,000	$320,000
Tax rate	4%	4%
Tax without exemption	$2,800	$14,000
Tax with exemption	$1,600	$12,800
Real rate without exemption	2.80%	2.80%
Real rate with exemption	1.60%	2.56%

Table 11.5: Effect of a homestead exemption on effective tax rates depending on the value of the property.

Property tax relief policies aim at enhancing equity and include the homestead exemption, which reduces the taxable value of a primary residence, and the circuit breaker, which provides tax relief based on income, particularly for lower-income households.

The homestead exemption provides an absolute reduction in taxable property values and is subtracted from the assessed value before the tax liability is calculated. Since it provides an absolute reduction in assessed value, lower home value see a higher percent reduction in taxes (Table 11.5). The homestead exemption applies to owner-occupied housing only and not to renters. The advantages are that it provides tax relief and improved vertical equity for low- to middle-income homeowners. However, it is neither helpful to renters who are often lower income households nor to cities with a large proportion of renters.

The circuit breaker provides rebate/credit on state income tax for high property taxes. It neither lowers the property tax bill nor directly affects local tax revenue. It can be applied to both, homeowners and renters, and is usually means-tested. The circuit breaker structure can be expressed as follows:

$$R = p \cdot (T - k \cdot I)$$

where R is the tax credit, p is a percentage specified by law and usually declining with income after a threshold, T is the property taxed paid, and $k \cdot I$ is the excessive tax burden specified as percent (k) of income (I). The property tax for renters can be estimated in order to apply the circuit breaker. Let us consider a simple example. Suppose that there is a tax credit of 50% on the excess burden if property taxes larger than 5% of income. In that case, $p = 0.5$ and $k = 0.05$. Thus, if the income is $30,000 and the property tax liability is $2,000, the income tax credit can be calculated as follows:

$$R = 50\% \cdot (\$2,000 - 5\% \cdot \$30,000) = \$250$$

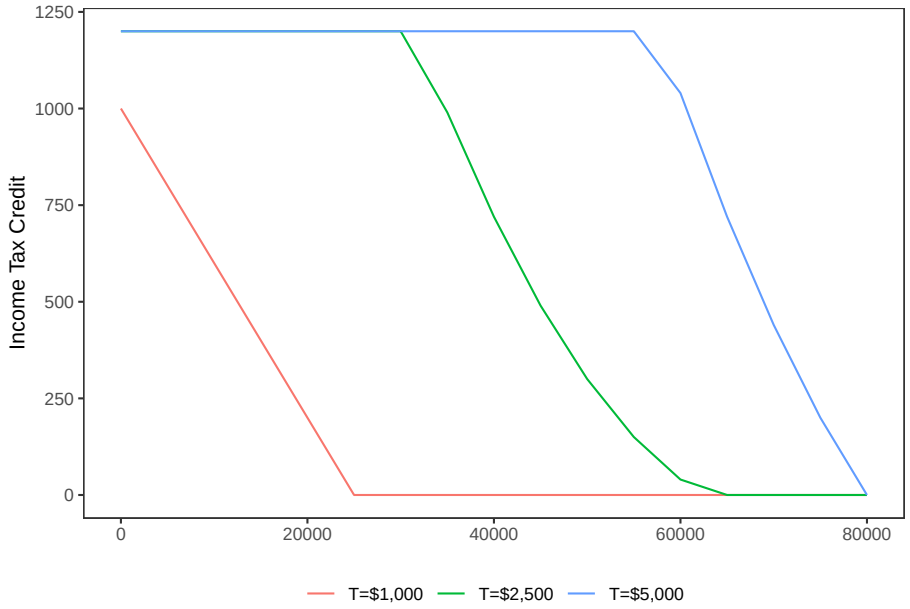


Figure 11.2: Illustration of the circuit breaker tax credit for the example in which the credit is limited to USD 1200 and the credit is reduced as income increases.

Let us consider a slightly more complex example. Consider the situation with a tax credit limit of $1,200. Suppose that the credit reduction of p is equivalent to 10 percentage points for every $5,000 over $30,000 of income and let $k = 4$.

The circuit breaker system offers several advantages and one notable disadvantage. One of the key advantages is that property tax relief can be specifically targeted to lower and middle-income homeowners and renters, making it an effective tool for addressing equity concerns. Additionally, the benefit can decrease as income increases, allowing for a more tailored approach to providing relief based on financial need. However, a significant disadvantage of the circuit breaker system is that households can only benefit if they file income tax returns. This limitation may exclude some eligible households that do not file, thereby reducing the reach of the program.

11.5 Evaluation Criteria

In evaluating property taxes from an equity perspective, effective tax rates serve as an important measure of tax burden relative to the ability to pay. Two possible approaches can be used: one involves calculating taxes divided by income, and the other divides taxes by market value. Horizontal equity refers to the principle

	Family A	Family B	Family C
Household size	8	2	2
Income	80,000	30,000	80,000
Adjustment factor	1.32	0.80	0.80
Adjusted income	60,606	37,500	100,000
Assessed value	250,000	100,000	250,000
Tax rate	2%	2%	2%
Tax liability	5,000	2,000	5,000
ETR (adjusted income)	8.25%	5.33%	5.00%

Table 11.6: Effective tax rates using equivalence-adjusted income

that individuals or properties with the same ability to pay should be taxed equally. However, horizontal inequities can arise due to inaccurate assessment values. Even with accurate assessments, horizontal inequities may still occur if the measure of ability to pay is based on income rather than property value. Vertical equity, on the other hand, addresses how tax burdens differ based on different levels of ability to pay (Table 11.6). Note that the adjustment factor is based in information of the U.S. Department of Housing and Urban Development. For residential properties, the tax burden typically falls on the homeowner. In the case of rental and business properties, the tax burden may be borne by landlords, tenants, owners, employees, or customers, depending on the prevailing market conditions.

In evaluating the efficiency of property taxes, key questions center around whether the tax distorts economic decisions and the extent of these distortions. The distortion of economic decisions can affect various areas. It can influence consumption decisions, particularly the choice between housing and other goods. It may also impact investment decisions, such as how much individuals or businesses invest in real estate. Additionally, business production decisions could be altered by the tax, affecting how businesses allocate resources. Finally, property taxes can influence location decisions for both individuals and firms, as tax rates may lead them to choose one location over another.

Adequacy in property tax evaluation focuses on the variations in revenue-raising capacity across different jurisdictions. These variations can be attributed to differences in wealth, with wealthier areas having a higher capacity to generate revenue. The presence of large industries can further increase a jurisdiction's ability to raise revenue. On the other hand, exemptions can undermine the capacity to generate revenue by reducing the taxable base. Elasticity refers to the potential for increasing tax revenue as incomes grow, without altering the tax base or the tax rate. This depends on the relationship between income growth, the market value of homes, and the method used for property assessments. A more elastic system can adapt to economic growth, ensuring that revenue increases naturally as the economy expands. Feasibility in property tax systems

is often influenced by the complexity of property assessment, which is considered the most challenging aspect. There is a trade-off between achieving horizontal equity and managing administrative costs. While property taxes are generally politically feasible, they are highly visible to taxpayers because they are often paid in a lump sum, either annually or semi-annually. Elected officials may hesitate to raise commercial property tax rates out of concern that higher taxes could drive businesses out of the area. In terms of stability, property taxes benefit from a broad tax base, which typically experiences little fluctuation over time. However, fluctuations in tax revenue can occur due to changes in property assessments, making assessment practices a key factor in the overall stability of the property tax system.